

The Verifiability of Recovered Memories of Abuse

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Abstract

Different ways of remembering sexual abuse - continuous memories, spontaneously recovered memories, and recovered memories following an active search - have been shown to differ in the availability of case-level corroborative evidence. The present study examined whether memory types in delayed sexual abuse statements also differ in terms of verifiability, a statement feature previously associated with differences between truthful and deceptive accounts. Victim statements from archival police files ($N = 140$) were classified according to how the abuse was remembered. Confirmatory analyses showed that statements based on continuous memories contained more verifiable details than statements based on recovered memories following an active search. Exploratory analyses indicated that details reported following an active search were more often disconfirmed than those reported in continuous or spontaneously recovered memories. Verifiability may extend beyond deception detection paradigms to capture systematic differences associated with memory retrieval and reconstruction processes in delayed sexual abuse reports.

Key words: Child sexual abuse, recovered memories, verifiability approach, false memory, statement validity assessment

Supplemental materials:

Preregistration, <https://osf.io/btp7y/overview>

Coding manuals and additional analyses, <https://osf.io/mkad9/overview>

General Audience Summary

Delayed reporting of child sexual abuse is common, but people differ in how they remember the abuse. Some people report that they always remembered the abuse, others describe memories that were retrieved spontaneously after a period of forgetting, and a third group reports memories that were recovered following an active search for possible past abuse, such as during therapy. Previous research has shown that these different memory types vary in the extent to which independent corroborative evidence is found, with recovered memories following an active search seldom being supported by strong corroborative evidence.

A complementary way to assess statement validity is through verifiability, that is, the extent to which a statement contains details that can in principle be checked. Verifiability has primarily been studied in deception detection research, where truthful accounts were shown to contain more verifiable details than fabricated accounts. The present study examined whether the three memory types differ in verifiable details. Using criminal case files, victim statements were classified by memory type and analyzed for *verifiable details* (i.e., elements such as witnesses or documents that could in principle be checked).

As expected, victim statements based on continuous memories of the abuse contained more verifiable details than statements based on recovered memories following an active search. An exploratory analysis showed that details reported following an active search were more often disconfirmed by other witness statements or other evidence than those reported in continuous or spontaneously recovered memories. These findings suggest that the amount of checkable information in a statement may not only be related to deception, but also to differences in how memories of abuse are recalled.

The Verifiability of Recovered Memories of Abuse

Disclosure of child sexual abuse is often delayed, influenced by shame, fear, loyalty conflicts, and concerns about not being believed (Morrison et al., 2018). Combined with the typically private, one-on-one nature of such abuse, the consequence of this delay is that physical and forensic evidence is frequently unavailable when cases are reported (Deferme et al., 2024; Read et al., 2006). As a result, assessments of abuse allegations often rely heavily on the content of the statement itself.

In some delayed disclosure reports, individuals claim periods of not remembering the abuse (Schooler, 2001). The validity of such recovered memories has been widely debated, as some may reflect false memories (Loftus, 1993). Although traumatic experiences are generally remembered well over time (Goldfarb et al., 2019), memory is reconstructive and vulnerable to distortion, particularly under the influence of misinformation (Loftus, 2005). When recovered memories are false, they may lead to false accusations and miscarriages of justice. Understanding the nature of such memories is therefore critical.

Different Ways of Remembering Sexual Abuse from the Distant Past

Delayed reports of sexual abuse can differ in how the abuse is remembered (Dodier & Patihis, 2021; Patihis & Pendergrast, 2019). Three types of memory reports in delayed disclosures of (sexual) abuse were proposed: continuous memories which were always accessible; spontaneously recovered memories, which emerge unexpectedly after a period of forgetting; and, recovered memories following an active search for possible past abuse, including but not limited to therapy (Ter Beek et al., 2026).

These memory types may differ in the likelihood of being true or false memories. Continuous memories and spontaneously recovered memories are typically assumed to involve spontaneous retrieval processes and hence, are more likely to be accurately retrieved (Lindholm et al., 2018). Recovered memories following an active search may rely more heavily on inference, expectation, and interpretation. Consequently, such recovered memories have been hypothesized to carry a higher risk

of being false, without necessarily implying intentional deception (Otgaar et al., 2025). Field studies show systematic differences in corroboration: continuous memories are most often corroborated, recovered memories following an active search least often, with spontaneously recovered memories occupying an intermediate position (Geraerts et al., 2007; Ter Beek et al., 2026).

The availability of corroborative evidence may partly depend on investigative decisions. An alternative approach to assess the validity of a statement is to examine the quality of the statement itself, for instance statement verifiability, a validated method from deception detection research (Nahari et al., 2014; Verschuere et al., 2021b). It is this method that we applied to examine the content of different memory types of sexual abuse.

Verifiability Assessment in Deception Detection

Research in deception detection has consistently shown that truthful statements typically contain more details than deceptive ones (Verschuere et al., 2023). Detail richness plays a central role in validated credibility assessment methods, including Criteria-Based Content Analysis (CBCA; (Volbert & Steller, 2014)). However, not all details may be equally informative.

Nahari and colleagues (2014) argued for the importance of distinguishing between *verifiable* and *non-verifiable* details. Verifiable details are elements that can in principle be checked, such as information that is documented or witnessed by identifiable sources. Laboratory studies in deception detection confirm that truthful statements typically contain more verifiable details than deceptive statements (Verschuere et al., 2021a). Note that in these studies the details are not actually checked, but rather it is the mere presence of verifiable details that distinguished the lies from the truths. However, these studies focused on deliberate deception whereas in delayed abuse reports the question is often whether memories are true or false. Importantly, false memories lack intent as they might be experienced as true memories. Potential differences in verifiable details in this context may arise from underlying memory processes rather than intentional deception. Whether verifiability is informative in this context has, as far as we know, not been examined.

Do Genuine and False Memories Differ in Verifiable Details?

The concept of verifiability originates from deception detection research but may also inform questions concerning the validity of autobiographical memories rather than the intentional fabrication of statements, including delayed reports of child sexual abuse. However, findings from deception research may not generalize to genuine versus false memories. Individuals reporting false memories typically experience them as genuine and may not strategically withhold details that characterizes deceptive accounts (Volbert & Steller, 2014). Moreover, false memories often integrate fragments of real experiences with schemas and expectations potentially resulting in detailed and coherent accounts (Hyman et al., 1995). Differences in overall detail richness between true and false memories may therefore be limited.

Empirically, findings from laboratory studies have been mixed. However, a recent scoping review identified the amount of detail reported as one of the most robust and consistently observed differences between true and false autobiographical memories (Wachendörfer & Oeberst, 2023). Supporting this conclusion, Wachendörfer and Oeberst (2024) found that true autobiographical memories contained more details than false memories.

These findings are broadly consistent with reality monitoring theory, which predicts that memories of experienced events are typically embedded in richer perceptual and contextual information than internally generated events (Johnson et al., 1988; Johnson & Raye, 1981). Such differences in contextual specificity may be reflected in the reporting of details that are potentially verifiable, such as identifiable persons, locations or activities. In addition, the availability of contextual information may help constrain or evaluate recollections, whereas its absence may allow greater reliance on inference and reconstruction. Case reports of retracted abuse allegations are consistent with this possibility, describing early recollections that only appeared implausible when attempts were made to situate the alleged events in time and place (Nierop & Van den Eshof, 2017; Ost et al., 2002).

Evidence from field studies further supports the possibility that memory-related processes may be reflected in statement characteristics. These studies examined statements from real cases involving alleged crimes and typically rely on corroborative evidence or judicial outcomes to approximate ground truth. Across studies, statements in cases judged likely to have occurred (e.g. based on medical, physical or scientific evidence; a confession by the accused person; or a guilty verdict) contain greater detail than those judged unlikely to have occurred (Amado et al., 2016; Sporer & Masip, 2024). Such studies likely include inaccuracies arising from both deception and memory-related processes and therefore capture the combined influence of these processes on statement content. Laboratory paradigms may not fully capture the complexity, autobiographical nature, and emotional salience of delayed abuse memories encountered in real-world forensic contexts. In legal decision making, considerable weight is placed on the characteristics of the statements themselves, particularly the richness of detail, when evaluating validity (Dreissen, 2025; Verschuere et al., 2025).

Taken together, theoretical accounts, field studies, and the importance attributed to statement characteristics in legal practice suggest that differences in how child sexual abuse is remembered may be reflected in the verifiability of victim statements. The present study therefore examined whether different memory types are associated with differences in verifiability. Using criminal case files, we tested whether victim statements based on memories recovered following an active search differ from those based on spontaneously recovered and continuous memories in the number of verifiable details reported.

Our preregistered hypotheses were:

(H1) Statements based on continuous memories will contain more verifiable details than statements based on memories recovered following an active search.

(H2) Statements based on spontaneously recovered memories will contain more verifiable details than statements based on memories recovered after an active search.

These hypotheses were tested following a preregistered analysis plan, with the number of verifiable details (sum score) as the primary outcome variable. Exploratory analyses examined a professional structured assessment of verifiability, in which trained raters provided a global guidelines-based judgment (using the same items), as a secondary outcome variable. Note that verifiability speaks to the extent to which details in a statement could potentially be checked. Laboratory studies do not allow to actually check the details that participants provided. Having actual police files, this study is the first to further explore the corroboration or disconfirmation of the reported verifiable details in subsequent police investigation, and relate this to the different memory types.

Method

The study was preregistered on the Open Science Framework (OSF; <https://osf.io/btp7y/overview>), where the coding manuals, coding sheets, and supplementary analyses are also available (OSF; <https://osf.io/mkad9/overview>). Permission to access and analyze the case files was granted by the Dutch Ministry of Justice and Security (reference number: 4803100). The research protocol was reviewed by the Ethics Committee of the Faculty of Psychology and Neuroscience at Maastricht University, which determined that formal ethical approval was not required. Data access, handling, and storage were conducted in accordance with strict governmental regulations.

Sample and Case Selection

The data were drawn from all case files submitted to the Dutch National Expert Committee on Complex Sexual Abuse cases (LEBZ) between 2008 and 2022. The LEBZ is a multidisciplinary advisory committee that evaluates the validity of statements in complex sexual abuse cases at the request of the public prosecutor, typically prior to a decision on prosecution (see Nierop et al., 2021; Ter Beek & Nierop, 2022 for detailed descriptions of the procedure). Consultation of the LEBZ is prescribed in cases involving recovered memories under Dutch guidelines (Nierop et al., 2021; Politie, 2024).

From the full pool of submitted cases ($N = 559$), a subset was selected based on predefined inclusion and exclusion criteria, described in detail elsewhere (Ter Beek et al., 2026). In brief, cases were eligible if they involved (a) recovered memories and/or (b) allegations in which the reported abuse had ended more than eight years prior to the formal police report. Cases were excluded if the complainant reported no memory of the alleged abuse, if there were strong indications of deliberate false allegations, if insufficient information was available to classify the memory type, or if objective coding could not be guaranteed.

After applying these criteria, the final sample consisted of 140 cases. The sample largely involved allegations of child sexual abuse, with substantial delays between the end of the alleged abuse and reporting. Of the victims, 86% were female, and in 95% of cases abuse reportedly started before the age of 16 years. The mean age at onset of abuse was 6.6 years ($SD = 5.4$), the mean age at report was 24.3 ($SD = 6.7$), and the mean delay between end of abuse and report was 12.4 ($SD = 6.5$). In 130 cases, a police investigation was available, allowing assessment of verification-related information. More detailed sample characteristics are described in a preprint based on the same dataset (Ter Beek et al., 2026).

Materials

The present study was based on a unique collection of police case files ($N = 140$). Each file contained one or more victim statements, derived from detailed investigative interviews. In most cases (93%; $n = 130$), the files also included materials from an extensive criminal investigation, such as witness statements, records of telephone or digital communication, and medical or psychological documentation. No new data was collected for the current study. Memory types were classified in an earlier study using this dataset (Ter Beek et al., 2026), and all subsequent coding was conducted based on the information contained in the case files. Owing to the sensitive nature of the materials and the data-handling restrictions imposed by governmental approval, the dataset was anonymized for group-level analyses and cannot be made publicly available.

Coding

Procedure

Coding of the police files was conducted by two pairs of coders either at a police facility or within the secure digital police environment. Memory-type classification was performed by two trained coders (MtB & AH) who were not involved in the cases they coded. Statements were classified into three memory types following a structured two-step procedure. First, coders determined whether the complainant reported a period of complete absence of memory for the alleged abuse. Cases without such a period were classified as *continuous memories*, whereas cases involving a period of reported complete forgetting were classified as *recovered memories*. Second, recovered memories were differentiated based on the manner in which recall occurred. Memories that returned suddenly and unintentionally, or following a single incidental trigger, were classified as *spontaneously recovered*. Memories that emerged gradually following suggestive influences or deliberate efforts to retrieve or interpret past experiences were classified as *recovered following an active search*. Importantly, active search was not limited to therapeutic contexts and could also involve influences outside therapy. Coding was guided by a structured manual and decision flow chart (OSF; Appendix 1: Coding Manual Memory Type; Appendix 2: Codebook Memory Type). Inter-rater reliability for the three nominal memory-type categories was $\kappa = .84$ (PABAK = .86). Detailed procedures and reliability estimates are reported elsewhere (Ter Beek et al., 2026).

A separate pair of coders (TDB and MH) evaluated the verifiable details. Coders were not blind to the study hypotheses; however, information about memory type was redacted from the victim statements such that coders were blind to memory classifications. This procedural separation minimized the risk of bias. Coder 3 (TDB) is a postdoctoral researcher in forensic psychology, and Coder 4 (MH) is a master's student in criminology. Neither had prior experience with criminal investigations of sexual abuse nor prior affiliation with the LEBZ.

Both coders received training from the first author (MtB), including a theoretical introduction to sexual abuse dynamics, memory processes, evidentiary assessment, and the identification of potentially verifiable details within historical child sexual abuse statements. Training included coding exercises using eight non-study cases before coding of the study cases began. Cases were coded by one of the two coders. A randomly selected subset was double-coded to assess inter-rater reliability. During routine coding, coders within the same team could discuss uncertainties. For the reliability sample, however, the coders worked independently and without discussion, and they had no access to each other's assessment.

Count-Based Coding of Verifiable Details (Preregistered)

The coding scheme for verifiable details was conceptually inspired by the Verifiability Approach (VA) developed in the deception detection literature (Nahari, 2018; Nahari et al., 2014). However, the present operationalization was adapted to the forensic context of historical sexual abuse cases and differs in at least two important ways from its original implementation.

First, the VA has primarily been applied to statements about past activities (e.g., alibis), in which concrete, externally observable actions are central. In contrast, historical sexual abuse cases typically involve substantial reporting delays, one-on-one situations, and limited opportunities for direct verification of the alleged acts. A narrow focus on observable activities of the sexual abuse itself would therefore insufficiently capture the types of information that could be linked to corroborative evidence in this context. Accordingly, verifiability was defined more broadly as the presence of statement details that could reasonably be subjected to investigation and that, if confirmed, would provide evidential support for the alleged abuse within a legal context. These elements are not restricted to direct witnesses or documented acts but also include witnesses to prior disclosure, witnesses of the context, witnesses or documentation of similar abuse by the same accused, contextual documentation (for instance, an e-mail confirming that the victim slept in bed with the accused), and abuse-specific knowledge that could plausibly be corroborated.

Second, given that our dataset consisted of extensive victim statements summarized in official police records rather than verbatim transcripts, word-by-word coding—as used in original VA—was neither feasible nor appropriate. Coding was therefore conducted at the level of distinct verifiable details as documented in the victim statement. Because police records typically include information considered relevant for investigative purposes, they provide a suitable basis for identifying potentially verifiable elements. This contextual adaptation is consistent with a prior application of VA by its developers, who modified both the unit of analysis and the operationalization of verifiability to suit a different applied setting, namely truth and lying about travel plans of airside participants (Jupe et al., 2017).

Each detail meeting the criteria of verifiability and potential corroborative value was counted once. The dependent variable, verifiable details, thus reflected the total number of such elements identified in a statement. For example, if a victim reported that a relative entered the room during an incident (potential identifiable witness) and that screenshots were retained of messages from the accused referencing the relationship (potential documented evidence), these elements were coded as two separate verifiable details and would lead to a sum score of 2.

Structured Verifiability Assessment (Exploratory)

In addition to the count-based VA measure, a guideline-based professional structured assessment of verifiability was conducted. Whereas the count measure quantified the *number* of verifiable details in this procedure, following a manual, the trained user could weigh in the potential evidential strength.

For the structured assessment, verifiable details were classified into two categories: (a) details that, if confirmed, would directly support the occurrence of the alleged sexual acts (e.g., documentation of the abuse, a witness to the abuse, a witness to a confession), and (b) details that would support contextual aspects or provide indirect corroborative evidence (e.g., disclosure

witnesses, witnesses of the context of the abuse, documentation of context of the abuse or indirect evidence). Across these categories, nine predefined items were evaluated (see Table 1).

Table 1

Different types of verifiable details in the context of delayed reports of sexual abuse

	Details related to sexual acts	Details related to context or indirect evidence
Documents	Documents that confirm the abuse	Documents confirm the context Documents that confirm other abuse by the same accused
Witnesses	Witness of the abuse Witness to whom accused confessed	Disclosure witness Witness of the context Person who witnessed or experienced abuse by the same accused
Knowledge		Victim has specific knowledge of the accused that can only be known when abused

Each of the 9 item was rated on a three-point scale (0 = *insufficiently verifiable*, 1 = *somewhat verifiable*, 2 = *highly verifiable*), reflecting the potential evidential strength that confirmation of the detail would have for the allegation if confirmed during investigation. Importantly, the overall classification of a case (*insufficiently*, *somewhat*, or *highly verifiable*) was not derived mechanically by summing item scores. Instead, trained coders applied predefined structured decision guidelines specifying that abuse-related details carried greater weight than contextual details and outlining how different combinations of items should be integrated (OSF; Appendix 3: Coding Manual Verifiable Details; Appendix 4: Codebook Verifiable Details). Using the earlier example, the presence of a potential direct witness to the abuse and documentary evidence referencing the relationship would result in two verifiable details in the count-based measure. However, in the structured assessment, these details would not contribute equally. A potential direct witness to the abuse would carry greater weight than contextual documentary evidence because it would more directly support the

occurrence of the alleged sexual acts. Conversely, a statement containing multiple contextual witnesses or indirect forms of corroboration could yield a relatively high count of verifiable details while still resulting in only limited overall verifiability under the structured decision rules.

Coding of Investigative Outcomes of Verifiable Details (Exploratory)

Finally, coders used the police files to determine whether the verifiable details mentioned by the victim had been investigated and, if so, whether these details had been confirmed, refuted or remained inconclusive. Because the focus was on relative differences in investigative outcomes of reported verifiable details across memory types, and coders were blind to the memory-type classifications, the same coders performed both coding steps.

Coding other File Characteristics

We developed a coding format including factors that could influence the extent to which details could be reported (e.g., age of the victim, duration of the abuse, frequency of the abuse), which were included as potential covariates in subsequent analyses. In addition, descriptive variables regarding the alleged victim, the number of accused persons, the relationship to the accused, and type of therapy were included.

Inter-rater reliability

To determine the inter-rater reliability of the number of verifiable details counted, the Intraclass Correlation Coefficient (ICC) was calculated on the basis of 14 cases, based on absolute agreement and single measurements (two-way random effects model). The ICC is suitable for this analysis because it involves numerical data counted by two independent coders. The ICC was 0.88, 95% CI [0.67, 0.96]. These estimates were obtained for a coding task that required identifying and counting multiple types of verifiable details in complex narrative statements. Reliability was assessed on independently coded cases without discussion, reflecting a conservative estimate of agreement.

To determine the inter-rater reliability of the structured assessment of verifiability, a Weighted Cohen's κ was calculated based on 14 cases. The assessment of verifiability concerns categorical (ordinal) data, where each case was rated on a three-level scale reflecting the degree of verifiability (e.g., insufficiently, somewhat or high verifiability). Weighted Cohen's κ takes into account the order of the categories and also considers the severity of the deviation, making it a suitable test. The Weighted Cohen's κ was .82, 95% CI [.61, 1.00]. This assessment required coders to weigh multiple context-specific aspects of the statement at both the item level and in the overall judgment, making it more complex than counting individual verifiable details.

The distribution across the verifiability categories was skewed, with a relatively large proportion of cases being classified as insufficiently verifiable. This skewed distribution reflects the nature of the data. Because Cohen's κ can be affected by such imbalanced category distributions, the Prevalence-Adjusted Bias-Adjusted Kappa (PABAK) was also calculated to provide a more robust estimate of agreement under these conditions. The percentage agreement was 92.9%, with the coders reaching a different final assessment in only one of the 14 cases. The corresponding PABAK was 0.86 and 95% CI [0.59, 1.00].

Analyses

Analyses were conducted within a Bayesian statistical framework to quantify evidence for differences across memory types. All Bayesian models were estimated in R (version 4.6.0), using the brms package (Bürkner, 2017). Weakly informative priors were specified to regularize estimates while avoiding undue influence on the posterior estimates. Models were estimated with 4 chains of 6,000 iterations each (2,000 warm-up), yielding 16,000 post-warmup samples. For count outcomes, results are reported as rate ratios, where values greater than 1 indicate a higher number of verifiable details in the comparison group. For regression models, effects were evaluated based on posterior estimates and 95% credible intervals (CrIs). For contingency table analyses and model comparisons, evidence for competing hypotheses was quantified using Bayes factors (BFs), which indicate the relative

likelihood of the observed data under the alternative hypothesis compared with the null hypothesis.

Bayes factors close to 1 indicate that the data provide little evidential value, as they are approximately equally compatible with both hypotheses (see, e.g. Lee & Wagenmakers, 2013; Wagenmakers et al., 2018). To ensure transparency, full details of all analyses, including prior specifications and model code, are provided in the supplementary materials (OSF; Appendix 5: Details of Analysis).

Results

Confirmatory Analyses

Association Between Memory Type and Verifiable Details (sum score; H1&H2)

First, we examined whether the three memory types differed in the number of verifiable details reported in victim statements ($M = 2.33$, $SD = 2.11$; $Mdn = 2$, $range = 0–10$). The distribution was positively skewed, with most cases containing relatively few verifiable details (89% reported 0–4 details; see Figure 1).

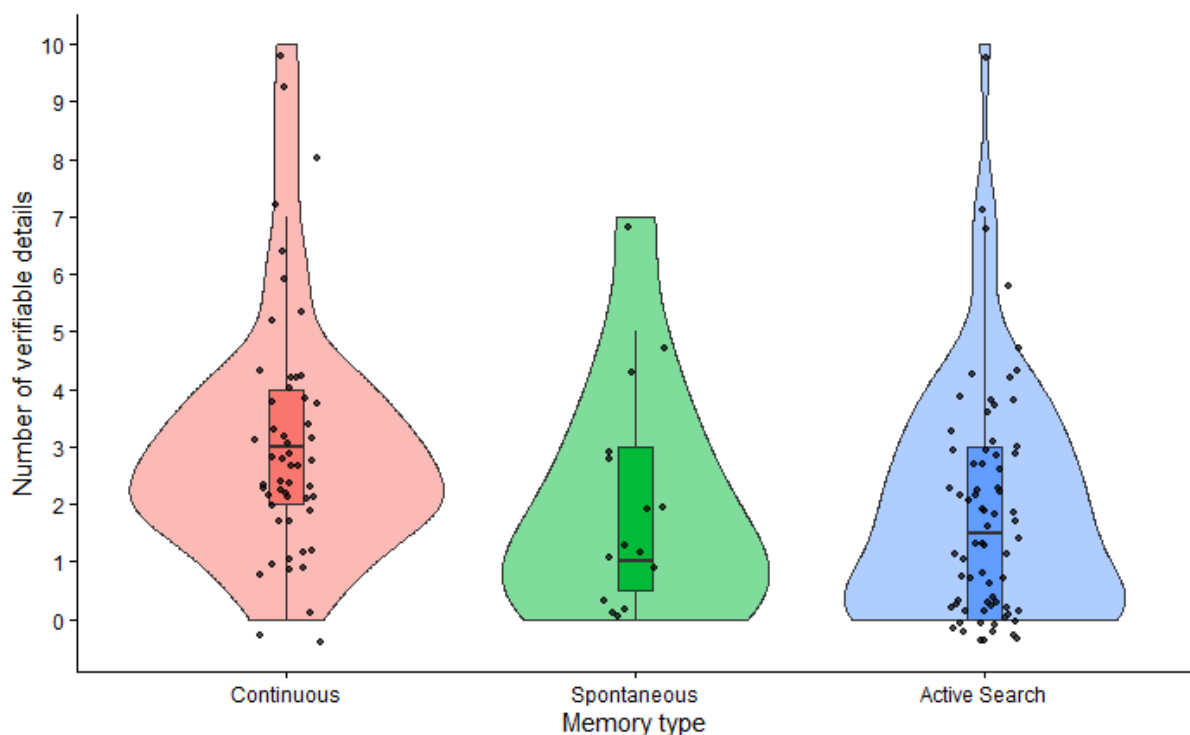


Figure 1

Verifiable Details per Memory Type

Note. Dots represent individual observations; the shape reflects the distribution of the data; central lines indicate medians and interquartile ranges. Continuous memories ($n = 53$), spontaneously recovered memories ($n = 15$), and memories recovered following an active search ($n = 72$).

Because the dependent variable was a skewed count variable with substantial overdispersion, the preregistered Mann–Whitney U test was not optimal. Instead, we used a negative binomial regression model with memory type (continuous, spontaneous, actively searched) as predictor. This deviation from the preregistered analysis plan was made because a negative binomial regression model provides a more appropriate and statistically valid approach for such data. Effects are reported as rate ratios, where values greater than 1 indicate more verifiable details in the comparison group. For transparency, results of the preregistered Mann-Whitney analyses – which mimic the current findings - are reported in the supplementary materials (OSF; Appendix 6: Supplementary Analyses).

Descriptive statistics indicated that continuous memories contained the highest number of verifiable details ($M = 3.04$, $Mdn = 3.0$, $SD = 2.10$, *range* 0 to 10), followed by spontaneous memories ($M = 2.00$, $Mdn = 1.0$, $SD = 2.07$, *range* 0 to 7), and recovered memories following an active search ($M = 1.88$, $Mdn = 1.5$, $SD = 2.01$, *range* 0 to 10).

Consistent with the directional hypothesis, continuous memories contained more verifiable details than memories recovered following an active search (ratio = 1.62, 95% CrI [1.16, 2.14]), indicating approximately 62% more verifiable details. In contrast, spontaneously recovered memories did not differ meaningfully from memories recovered following an active search (ratio = 1.07, 95% CrI [0.59, 1.65]).¹

Finally, robustness analyses excluding five outliers (i.e., deviated 2 SDs from the M) yielded a similar pattern of results, indicating that the findings were not driven by extreme values (OSF; Appendix 6: Supplementary Analyses).

¹ Frequentist negative binomial regression analyses yielded the same pattern of results (continuous vs active search: ratio = 1.62, $p = .006$; spontaneous vs active search: ratio 1.07, $p = .967$; see OSF; Appendix 6: Supplementary Analyses).

In line with Otgaar et al. (2022), we defined a difference of one verifiable detail as the smallest effect size of interest. This threshold was specified *a priori* in the preregistration. Although a single additional verifiable detail may appear modest in quantitative terms, it may carry substantive implications in the context of sexual abuse cases. Under Dutch criminal procedure, a conviction cannot be based on a single source of evidence; if the victim's statement is deemed reliable, it must be supported by at least one independent piece of evidence (Art. 342(2) Dutch Code of Criminal Procedure). Consequently, confirmation of a single verifiable detail may meaningfully strengthen the evidential position of a case.

To assess whether the observed differences reached this threshold, posterior estimates were examined for the preregistered comparisons using the regions of practical equivalence (ROPE) procedure. For the comparison between continuous memories and recovered memories following an active search, the posterior mean difference exceeded one verifiable detail ($M = 1.16$), with 66% of the posterior distribution lying above this threshold.

In contrast, for the comparison between spontaneous recovered memories and recovered memories following an active search, the posterior mean difference did not exceed the smallest effect size of interest of one detail ($M = 0.16$), and only 6% of the posterior distribution lay beyond this threshold, indicating little evidence for the practical relevance for the effect.

Exploratory Analyses

Association between Memory Type and Structured Verifiability Assessment

As a complementary exploratory analysis, we examined an alternative operationalization of verifiability based on a professional structured overall judgment. Because verifiability was measured on an ordinal scale (insufficient, somewhat, high), a Bayesian cumulative ordinal regression model was fitted as the primary analysis to assess whether statements in one memory group were generally more likely to receive higher verifiability ratings than those in another.

A Bayesian cumulative ordinal regression indicated that, relative to memories recovered following an active search, statements based on continuous memories showed a higher probability of being classified in higher verifiability categories ($b = 1.28$, 95% CrI [0.64, 1.95]). Statements based on spontaneously recovered memories also showed a higher probability of being classified in higher verifiability categories relative to recovered memories following an active search, although the estimated effect was smaller and substantially more uncertain ($b = 0.79$, 95% CrI [-0.31, 1.90]). Descriptive distributions of verifiability across memory types are presented in Table 2.

Table 2

Memory Type and the Verifiability of the Statement

Verifiability	Continuous Memories ($n = 53$)	Spontaneous Recovered Memories ($n = 15$)	Recovered following an active search ($n = 72$)
Insufficient	18.9% (10/53)	46.7% (7/15)	48.6% (35/72)
Somewhat	43.4% (23/53)	6.7% (1/15)	38.9% (28/72)
High	37.7% (20/53)	46.7% (7/15)	12.5% (9/72)

As a complementary analysis, one-sided Bayesian contingency table analyses were conducted for preregistered comparisons. For continuous versus recovered memories following an active search, the data were more likely under the hypothesis that statements in the continuous group were more verifiable than those in the recovered following an active search group than under the null hypothesis that there is no difference ($BF_{+0} = 415.8$). Similarly, for spontaneous recovered versus memories recovered following an active search, the data were more likely under the directional hypothesis than under the null hypothesis ($BF_{+0} = 71.81$). This latter comparison contrasts with the ordinal regression results, which were associated with greater uncertainty, possibly reflecting differences in model assumptions and the small size of the spontaneously recovered group.

Investigative Outcomes of Verifiable Details

When assessing the validity of a statement, it is of interest whether the details are supported or refuted. Unlike laboratory research, the present study provides a unique opportunity to examine the investigative outcomes of reported details. Of the 130 cases investigated, 31 did not contain any verifiable details. In the remaining 109 cases, verifiable details were reported and could be evaluated. Verifiable details that had not been investigated were disregarded. In total 220 verifiable details were investigated. Table 3 shows the relationship between memory type and investigative outcome (disproved, confirmed, unclear).

Table 3***Investigation Outcomes of Verifiable Details by Memory Type***

Investigation outcome	Continuous Memories <i>N</i> = 121	Spontaneous Recovered Memories <i>N</i> = 21	Recovered following an active search <i>N</i> = 78
Disproved	24.8% (30/121)	33.3% (7/21)	60.3% (47/78)
Confirmed	59.5% (72/121)	28.6% (6/21)	34.6% (27/78)
Unclear	15.7% (19/121)	38.1% (8/21)	5.1% (4/78)

*Note: Percentages reflect the distribution of investigated Verifiable Details (*N* = 220) across memory types. Each detail constitutes a separate unit of analysis.*

A clear pattern emerged in the proportion of details that were disproved. Details derived from memories recovered following an active search were most often disproved by investigation (60.3%) compared with substantially lower proportions in the continuous (24.8%) and spontaneous recovered memory group (33.3%). Thus, details from actively searched recovered memories were markedly more likely to be disproved.

Consistent with this pattern, a Bayesian multinomial logistic regression indicated that continuous memories had a lower posterior mean predicted probability of being disproved than memories recovered following an active search (-32.9 percentage points; 95% CrI [-45.3, -19.5]) and a

higher posterior mean predicted probability of being confirmed (24.0 percentage points; 95% CrI [10.4, 37.0]. Continuous memories also had a higher posterior mean predicted probability of yielding an unclear outcome than memories recovered following an active search (8.9 percentage points; 95% CrI [0.9, 16.9].

In comparison, spontaneously recovered memories had a lower posterior mean predicted probability of being disproved than recovered memories following an active search (-21.9 percentage points; 95% CrI [-42.2, -.11] and a higher posterior mean predicted probability of resulting in an unclear outcome (28.9 percentage points; 95% CrI [9.9, 49.8]. The difference in confirmed outcomes between memories recovered following an active search and spontaneously recovered was uncertain (7.0 percentage points; 95% CrI [-14.8, 26.6]).

Overall, continuous memories showed more favorable investigation outcomes than recovered memories following an active search, whereas spontaneously recovered memories were primarily characterized by fewer disproved and more unclear outcomes.

Bayesian contingency table analysis provided additional support for an association between memory type and investigative outcome across all three categories ($BF_{10} = 454582$). Given that an unclear outcome is less informative with respect to the evidential value of a reported verifiable detail, additional analyses focused on determined outcomes (confirmed vs. disproved). These analyses yielded strong evidence that details from memories following an active search were more often disproved than those from continuous memories ($BF_{10} = 18568$). In contrast, the comparison between spontaneously recovered memories and memories recovered following an active search yielded only limited evidence for a difference ($BF_{10} = 1.46$), likely reflecting the relatively small size of the spontaneous recovered group.

Additional exploratory and robustness analyses are reported in the supplementary materials. First, robustness analyses examined whether the observed differences in verifiable details remained after adjustment for abuse-related characteristics that differed across groups in a previous analysis of

the same dataset (Ter Beek et al., 2026). Results were substantively unchanged, indicating that the observed differences were not attributable to these variables (OSF; Appendix 6: Supplementary Analyses). Second, exploratory analyses examined whether the types of verifiable details differed across memory groups. Given sparse data across several categories, particularly in the spontaneously recovered memory group, these analyses were considered descriptive (OSF; Appendix 7: Distribution and Comparison of Verifiable Detail Types).

Discussion

In this study, we examined the verifiability of recovered memories of abuse using criminal case files. Statements based on continuous memories contained more verifiable details than those based on memories recovered following an active search, whereas differences between spontaneously recovered memories and memories recovered following an active search were negligible. Moreover, details reported in memories recovered following an active search were more frequently contradicted by external evidence. These findings extend verifiability research beyond laboratory deception detection paradigms by suggesting that differences in verifiable details may emerge in the absence of intentional deception. This is consistent with theoretical accounts proposing that variation in reported detail may reflect underlying memory processes rather than solely strategic reporting (Johnson & Raye, 1981).

Our findings align with field research suggesting that discriminative differences in statement characteristics may be stronger in ecologically valid interpersonal violence contexts than in laboratory paradigms (Amado et al., 2016; Sporer & Masip, 2024). In contrast, a recent survey-based study found no differences in verifiability between true and fabricated sexual abuse accounts (Bošković et al., 2025). However, that study involved only low- to medium-impact sexual offenses. More broadly, analogue studies often rely on brief statements, limited retrieval periods, and minimal social or legal consequences. By contrast, delayed sexual abuse reports in legal settings often concern highly impactful experiences, involve lengthy retrieval and reporting processes, and may lead to criminal

prosecution. Consistent with this interpretation, verifiability effects have been shown to increase when individuals expect their statements to be checked against external evidence (Palena et al., 2021). Laboratory studies may not sufficiently capture the contextual and motivational characteristics of real-world abuse disclosures, highlighting the importance of further research in ecologically valid forensic settings.

Beyond the number of verifiable details, a practical challenge concerns how such information should be interpreted when evaluating statement validity. Translating the quantity of verifiable details into accurate truth judgments is not straightforward and prone to substantial variability between evaluators (Gálvez-García et al., 2025). The present findings suggest that the evidential value of verifiable details may depend not only on their quantity, but also on whether such details are subsequently confirmed or contradicted by external evidence. Some statements contained numerous verifiable details that could be investigated, yet investigation showed that these details were incorrect or contradicted by external evidence. Conversely, statements containing relatively few verifiable details may still provide strong support when the available details are confirmed. Importantly, disconfirmed details occurred most frequently in statements based on recovered memories following an active search. This pattern is consistent with the possibility that accounts involving substantial reconstruction or suggestive influences may rely more on inference, schemas, and general knowledge, thereby increasing the likelihood that reported details do not correspond to external reality. From this perspective, the diagnostic value of verifiable details may lie not only in their quantity, but in the combination of the number of verifiable details reported and whether those details are subsequently confirmed or contradicted during investigation, rather than relying on the quantity of details alone.

The present findings also help to contextualize earlier results based on the same dataset (Ter Beek et al., 2026)², which showed lower rates of case-level corroboration for memories recovered following an active search. One possible explanation for such findings is that allegations involving recovered memories may receive less extensive investigation due to skepticism toward these cases. However, the present analyses suggest that differences in corroboration are unlikely to be explained by differences in investigative effort. Statements based on memories recovered following an active search not only contained fewer verifiable details, but the details that could be investigated were also more frequently contradicted by external evidence. This indicates that investigations did take place in these cases, but that the reported details more often proved inaccurate. These findings suggest that lower corroboration rates are at least partly related to differences in the content and accuracy of the statements themselves.

Together, the findings suggest that examining verifiable details provides a complementary approach to evaluating the validity of statements, particularly in contexts where corroborative evidence is limited. However, the evidential value of such details may depend not only on their presence or quantity, but also on whether they are confirmed or contradicted during investigation. This interpretation aligns with recent meta-analytic findings showing that most verbal cues provided limited discriminative value, whereas correspondence with external evidence is more informative (Vredeveltdt et al., 2026).

In sum, statements based on continuous memories contained more verifiable details than those based on recovered memories following an active search, whereas the two recovered memory groups showed similar levels of verifiable details. Crucially, details reported following an active search were more often contradicted by external evidence. These findings suggest that verifiability may be relevant beyond deception detection contexts, although its interpretation may depend not only on

² The previous study examined corroboration at the case level and included all forms of corroborative evidence. The present analyses focused specifically on whether verifiable details reported in victim statements were subsequently corroborated or contradicted.

the quantity of verifiable details reported, but also on whether such details withstand investigative scrutiny.

Author Contributions

Conceptualization (BV, CdR, HO, LR, MtB, TDB); Data curation (AH, MH, MtB, TDB); Formal Analysis (BV, HO, MtB, PR); Funding Acquisition (BV, HO, MtB); Methodology (BV, HO, MtB), Project Administration (HO, MtB); Resources (BV, HO, MtB, TDB); Supervision (BV, HO); Writing-Original Draft (BV, HO, MtB); Writing – Review & Editing (BV, CdR, HO, LR, PR).

Conflict of Interest

The authors declare that they have no financial conflicts of interest. Four of the authors are currently affiliated or have previously been affiliated with the expert committee that advised on the police files included in this study. None of the affiliated authors coded the verifiable details and none analyzed any case in which they had been previously involved. The authors affirm that the research was conducted independently, and that no committee involvement influenced the study design, data analysis, or interpretation of the findings.

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